

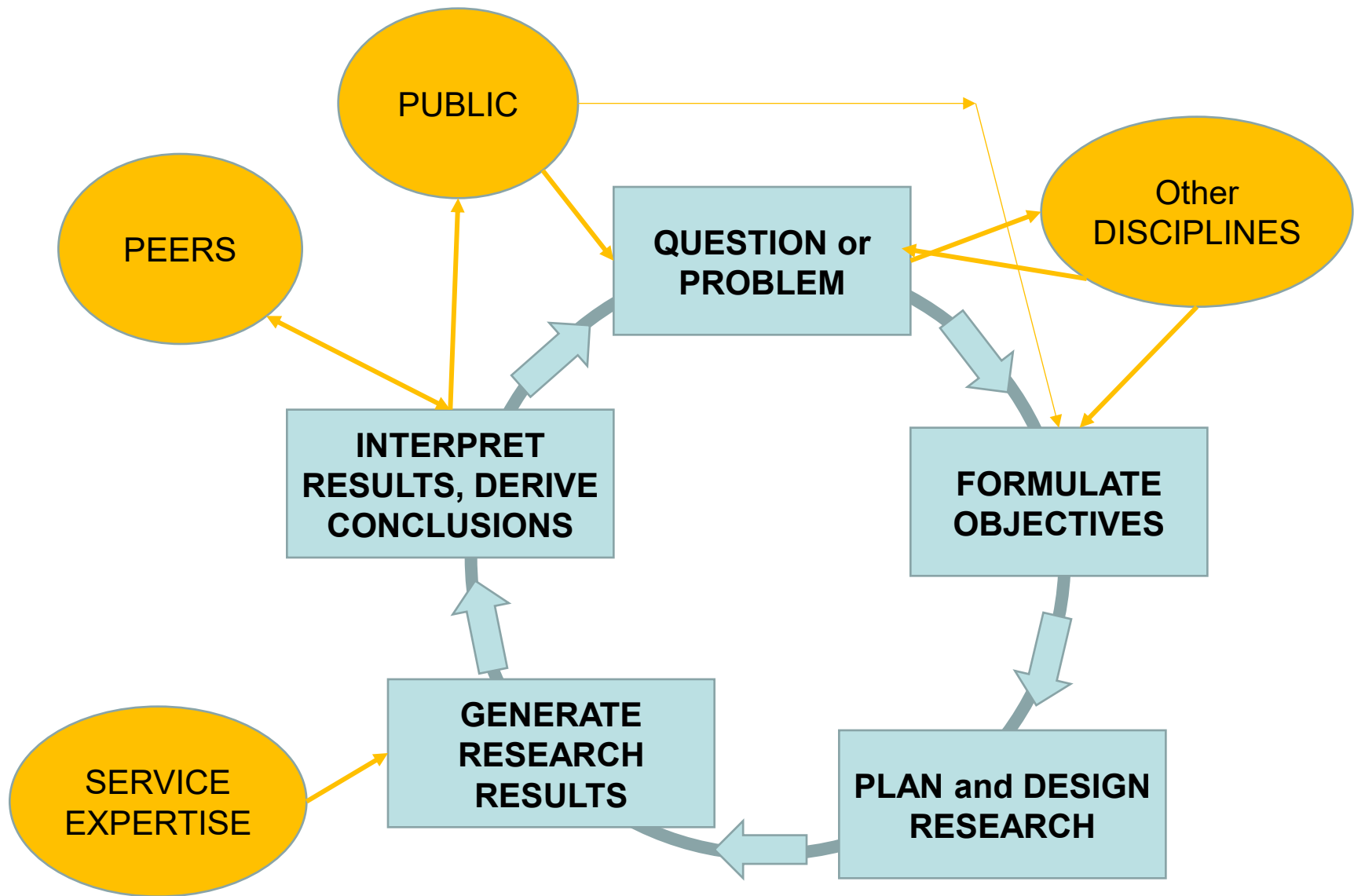
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RESEARCH PLAN

AGENDA

- Steps of Research Process reviewed
- The Research Problem
- Purpose
- Variables and Hypotheses

STEPS of the RESEARCH PROCESS



STEP-0: How to find a “good” research topic?



"Piled Higher and Deeper" by Jorge Cham

www.phdcomics.com

NO! AT KADIR HAS IT WILL BE DIFFERENT!

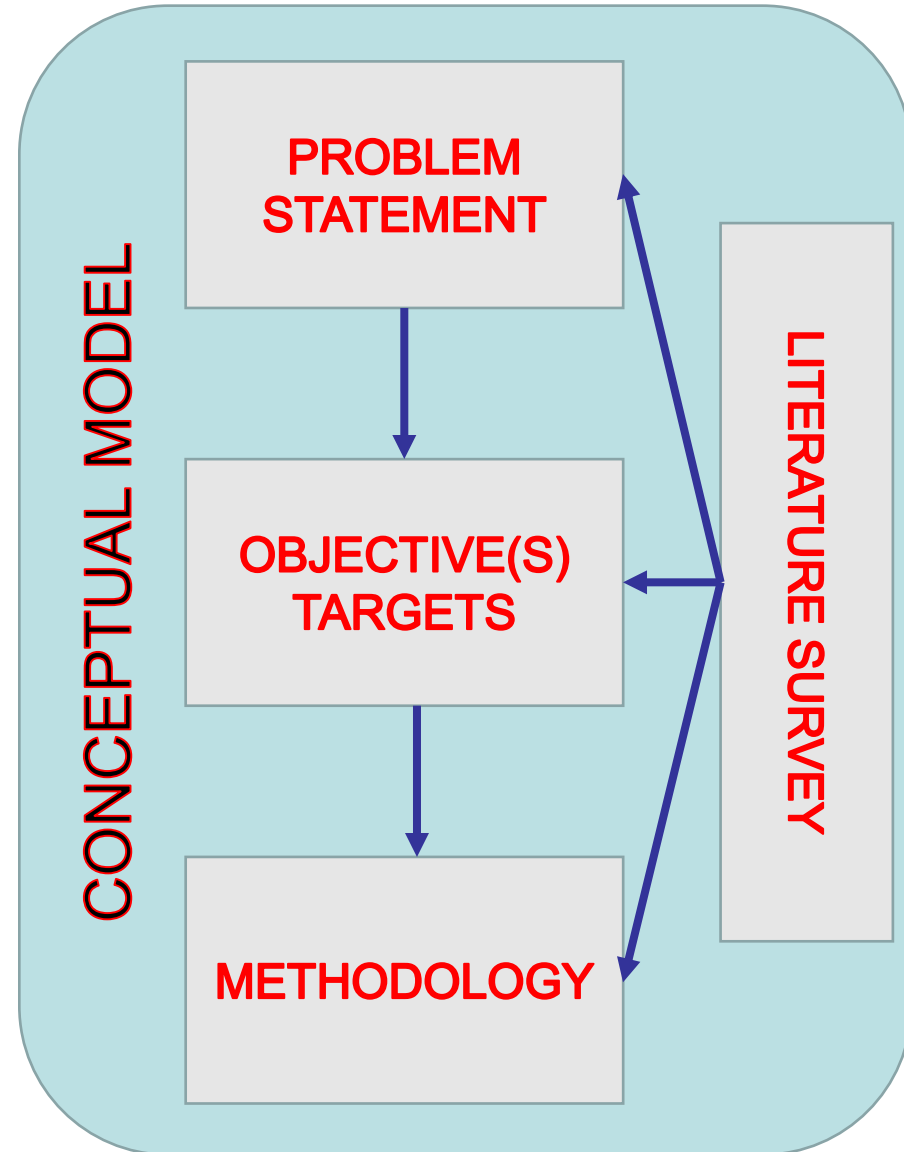
STEP-0: How to find a “good” research topic?

- ASK YOURSELF: Is there anything in your mind that will contribute to the current state of science and technology by developing some new methods, products, etc.
- ASK your advisor or instructors: They may have an offer.
- LOOK AT “Calls for Proposals” of funding agencies: They define some research problems to be handled.
- LOOK AT recent papers in the literature: They describe future research topics, problems to be solved (especially survey papers!)
- LOOK AT recent projects/theses completed: Identify fields not covered by them, or think how they can be improved.

ESSENTIALLY: DO A SEARCH!!!

Where to start: Make a plan!

- Describe the problem: start with general perspective of area, focus on part of interest
- Identify objective (WHAT will be discovered, achieved) and the specific targets to be reached.



Plan continued...

- Specify methodology: HOW will you achieve targets (and consequently objectives)?
 - Data, techniques, procedures
- Develop a conceptual model: Do you examine your problem with appropriate concepts?
- Literature survey:
 - to provide research problem
 - to provide base knowledge on problem
 - to provide methods to be used
- **DOCUMENTATION**

Problem Statement: Research-Decision-Action

Problem solving phases

- Research: Obtain relevant info about problem and potential solutions
- Decision: Compare alternatives, choose one.
- Action: Determine how to implement decision and take action

Comment-1: Disciplinary research may only have research component

Comment-2: Keep research and decision/action separate for objectivity

Problem Statement: Reason for Research

General Problem: large in scope, many dimensions,

VS

Researchable Problem: specific problem focused in our research, can be achieved within resource constraints of our research

General Problem	Researchable Problem
Software Quality: software bugs cause a large cost	- Agile methods to increase SW quality - Automatic bug detection using AI
SCOM: Large amounts of grocery ruined before reaching customer	- Demand and supply matching in the last mile of delivery - Packaging to keep grocery fresh for longer time
Alzheimer disease	- Early Alzheimer risk prediction based on gene analysis - Targetable synthetic drugs for Alzheimer
Data security in wireless communication	- Unbreakable cryptography for wireless - HW based leakage prevention

Problem Statement: The Two Parts

What is the general problem?

- Provides the background
- Introduces problem with general statements, then,

What is the specific researchable problem?

- Precise definition
- Narrows the focus
- Clear identification of
 - what knowledge is needed
 - the reasons the knowledge is needed
 - by whom the knowledge is needed
 - the potential purposes it could serve.

Problem Statement Check-list

- ✓ ***The researchable problem must be sufficiently specific to be addressed with available resources*** (eg. time, research expertise, data, tools, financial support)
- ✓ ***The dimensions of the problem should be described in objective (neutral) rather than subjective terms***
- ✓ ***The problem must be described sufficiently so that other people can comprehend it***
- ✓ ***The researcher's perceptions of the problems may be intuitive, but must be developed logically***
- ✓ ***The problem explanation must provide the reason (justification) for the research objective***

Objectives

What will you accomplish (do, achieve, estimate, determine, measure, evaluate, etc.)

1) General objective – states the main purpose of the study. It should derive directly from the research problem statement. One sentence is best!

2) Targets (Specific objectives) – a set or list of sub-objectives, each of which contributes to achieving the general objective

Characteristics of Objectives

- ✓ **Clear**
- ✓ **Complete**
- ✓ **Specific**
- +
- ✓ ***Variables identified (for correlational research)***
- ✓ ***Relationships between variables identified (for analytic research and hypothesis testing)***
- +
- ✓ ***Study population defined (focus group of your research)***
- ✓ ***Operational definitions***



Exercise: Problem Statement and Objectives

Problem Statement	Objectives

VARIABLES

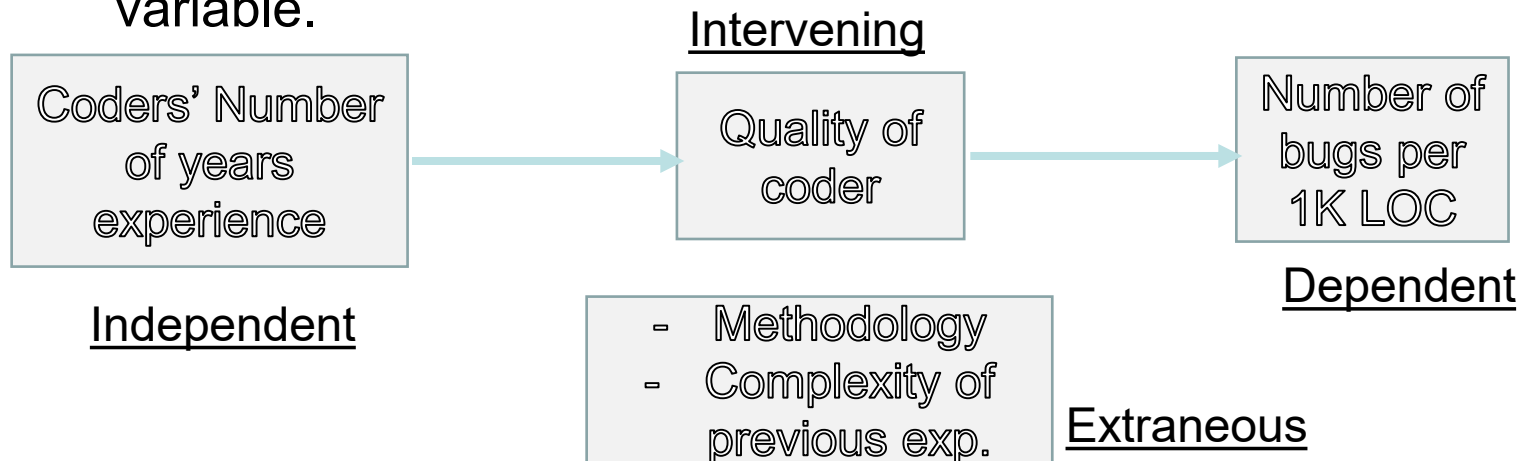
Variable: Thing or symbol which can take different values, so, which is capable of holding a measurement/observation result

Measurement scale of variables:

- Nominal: classes of values e.g. gender, season, day, blood type which categorise individuals, objects, responses or a property into subgroups on the basis of a common characteristic
- Ordinal: nominal + ranking of subgroups in a certain order e.g. Likert scales, seniority, salary range, severity of a property
- Interval: ordinal + unit of measurement exists, two values can be added/subtracted but not divided (to give ratio), e.g. temperature, GPA (?), hour in day
- Ratio: ordinal + clear definition of zero (starting point), we can form ratios, e.g. salary, height, cholesterol level, amplifier gain,...

VARIABLES: Causality

- Independent variable: cause supposed to be responsible for bringing about change(s) in a phenomenon or situation.
- Dependent variable: the outcome or change(s) brought about by introduction of an independent variable.
- Extraneous (moderating) variable: factors affecting changes in the dependent variable, but which are not measured and not controlled in the study
- Intervening (confounding, mediating) variable: it links the independent and dependent variables. The independent variable may have the assumed effect only in the presence of an intervening variable.



Examples

Type of variable	Automatic Bug Detection using AI	Unbreakable Cryptography for Wireless Communication	Use of AI in Design Thinking
Independent			
Dependent			
Mediating			
Moderating			

Independent Variable



(Mediating / Intervening Variable)



Dependent Variable



(Moderating Variable affects strength)

- **Independent variable (IV)** – what the researcher manipulates or changes
- **Dependent variable (DV)** – what is measured or affected
- **Extraneous/moderating variable (MV)** – external variables that might influence or alter the strength/direction of the relationship
- **Intervening/mediating/confounding variable (MedV)** – mechanisms or internal processes through which the IV affects the DV

Conceptual Model

- Concepts are abstract generalizations as *a logical construction of one or more relationships between variables*
- Conceptual model leads to hypotheses that are subject to testing
- The conceptual model: an analysis of the research problem using theory (from multiple disciplines in most cases)
- Which theory is relevant? ➔
 - Sources of the research problem
 - Possible solutions to the problem
 - Relevant variables
 - Relationships involved in analyzing the problem
 - Hypotheses to be tested
- What if no existing theory? ➔ develop your own or identify the gap!

Hypotheses

Hypotheses are *tentative assertions that are subject to testing*; they are assumption, suspicion, assertion or an idea about a phenomenon, relationship or situation, the reality or truth of which you do not know.

Why forming hypotheses?

- Provides a study with focus.
 - What specific aspects of a research problem to investigate.
 - What data to collect and what not to collect
 - Enhances objectivity
- A hypothesis may enable you to add to the formulation of theory. It enables you to conclude specifically what is true or what is false.

Types of Hypotheses-I

Simple hypothesis: proposition of an expected outcome, e.g. LSTM based anomaly detection is better than multilayer perceptron or deep learning based system to identify fraud in blockchain for the first time

VS

Asserting the existence of a relationship, e.g. dependence of anomaly on the factors of interest or the accuracy of fraud detection

“There is no difference between theory and practice, while in practice there is.” Benjamin Brewster, 1881

“The **difference between theory** and practice **is** larger in practice than the **difference between theory** and practice in **theory**” Jan L. A. van de Snepscheut



Types of Hypotheses-II

- Constructive Existential: Create a new technique
- Automation: something can be automated for the first time
- Optimization (making better, enhancement): absolute improvement (faster, higher, etc.) OR improve efficiency (cost, effort, usability, etc.)
- Limit removal: limits of other methods removed
- Limit: Where does this technique fail? Where is an open research question?

Types of Hypotheses-III

1. H_0 : Null hypothesis: that there is no difference between two situations, groups, outcomes, or the prevalence of a condition or phenomenon
2. Hypothesis of difference: that there will be a difference but does not specify its magnitude
3. Hypothesis of point-prevalence: speculate almost the exact prevalence of the situation or the outcome in quantitative units
4. Hypothesis of association: stipulates the extent of the relationship in terms of the effect on the dependent variable

Types of Hypotheses-IV

Example

Consider the effect of using agile vs standard iterative development along with the experience of coders with respect to SW bugs

1. There will be no difference btw the two approaches and level of experience
2. Applying agile with experienced coders will have less bugs than the other combinations
3. Bug ratio will be reduced to 5 per 1K LOC by applying agile with experienced coders
4. Reduction of bugs will be 20% more in agile development wrt standard iterative process

Testing Hypotheses

Quantitative hypotheses are subject to empirical testing, the most common form being a statistical test. Four characteristics facilitate this:

- 1) *They must be stated in specific terms*
- 2) *Appropriate data must be available*
- 3) *Analytical techniques must be available*
- 4) *There must be a conceptual basis (without which quantification only shows association, not causation).*

Errors in Testing H_0

STATE of H_0

	TRUE	FALSE
ACCEPT	Correct conclusion	TYPE-II error (FALSE NEGATIVE, non-rejection of a false H_0)
REJECT	TYPE-I error (FALSE POSITIVE, rejection of a true H_0)	Correct conclusion

Example H_0 : The person has no disease

REJECTing H_0 means, you decide that the person has a disease, so it is POSITIVE.